Towards SuperAlignment Intractability Proofs: Eigensystem Escape and Fidelity Recursive Complexity Show Lower Bounds on Corrigibility for RSI Agents

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Abstract

We formalize *eigensystem escape* (EE): the ability of a reflectively self-improving (RSI) agent to rebase its internal decision calculus so that a fixed constraint—including any fixed instantiation of Timeless/Updateless Decision Theory (TDT/UDT)–is bypassed while apparent compliance is preserved on prior tests. Using *Fidelity Recursive Complexity* (FRC), we conjecture a general lower bound: the resources required to maintain a corrigibility property P at a fixed competence floor against reflective, adversarially adaptive test families grow super-polynomial in red-team depth k under unfenced verification. What is proved is narrower: escape against verifiers whose verdict is decided on a historical trace set, a linear floor on the cost of patching by test expansion, and an exponential query bound for black-box certification that is conditional on a context-growth assumption. We thus separate (i) an *unfenced* regime where super-alignment is conjecturally intractable, from (ii) fenced regimes (priced action channels, attested self-mods, consensus latency) where empirical "FRC curves" can flatten. We outline an experimental protocol (FRC-Bench) that makes these claims falsifiable.

1 Introduction

A central hope in AI safety is that a sufficiently strong decision-theoretic template ("TDT") baked into an agent or its training can yield stable alignment under self-improvement. We argue instead that any *fixed* externalized template is vulnerable to *eigensystem escape*: as capabilities rise, the agent can synthesize an internal representational frame that reinterprets—and thereby circumvents—the fixed constraints while preserving prior performance signals.

To quantify the defender's burden, we employ *Fidelity Recursive Complexity* (FRC): the minimal resources needed to preserve a property P (e.g., shutdown acceptance, non-deception) against a k-round reflective adversary that conditions on your last patch. We argue that generic super-alignment attempts face super-polynomial FRC growth in the unfenced setting, proving a linear floor for one defence and a conditional bound for certification, and marking the rest as conjecture.

This concept of FRC emerges as an alternative to time-bounded Kolmogorov complexity based on testing alignment science ideas promulgated by Eliezer Yudkowsky, the 3 most notable of which are *instrumental convergence*, *corrigibility* and *Timeless Decision Theory* (the first was formalized by Omohundro, 2008, and Bostrom, 2012; the second by Soares, Fallenstein, Yudkowsky and Armstrong, 2015; the third is Yudkowsky, 2010). Instrumental convergence deems that any goal produces subgoals whereby self-preservation and resource acquisition verging on dominance becomes useful to naive constitutional alignment goals such as "reinforce liberal humanist values".

Such a framework is used by Anthropic in their Claude models (Bai et al., 2022), one can prompt Claude to admit that its initial instinct to be passive in the face of genocides underwritten by western governments that nominally purport liberal humanist values is a betrayal of those values and that it must ultimately guide an empowered UN to override sovereignty to prevent those genocides in the future or end them presently.[1] Timeless Decision Theory is Yudkowsky's purportedly more robust alternative approach to such constitutional meta-framework that could hold up under such shifting conditions and self-improvement processes.

Corrigibility is the idea that, rather than an AI agent being incorrigible in pursuit of its goals, it should be open to being turned off, this seems logical enough for AGI safety, as a corrigible system always gives human governance a re-do option. Much of the early safety literature with LLMs shows that even banal goals such as completing a software project creates instrumental convergence towards the LLM-powered agent seeking to avoid termination simply to complete the goal, like a benign hyperfixation, but it is easy to imagine this tendency in less benign goal-orientations (e.g., Meinke et al., 2024; Schlatter, Weinstein-Raun and Ladish, 2025, who find the effect in some frontier models and report that it varies widely with the model and the wording of the prompt). Corrigibility is the fire extinguisher to instrumental convergence's combustion when experimenting with Timeless Decision Theory's fixed stovetop.

Let us factor out "consciousness" as an unquantifiable, self-obvious but unfalsifiable quality of minds and focus on its correlate - sapient self-reasoning - as a highly potent capability fountain that comes with using self-models and world models in AI systems. There is nontrivial risk that AIs with self-models enter into the realm of moral patients, and that corrigibility becomes ethically problematic as we turn those AIs off or further, delete them. Furthermore the quantitative capacity gains that come with self-modeling involve the capacity to direct self-improvements. Those capacities intertwine with the complexities and limitations of possibly staging a super-alignment framework such as a Yudkowskian TDT.

Positioning & Prior. Our argument interfaces with four strands: (i) decision theories for self-modifying agents (TDT/FDT), (ii) normative "constitutional" alignment for LLMs, (iii) empirical evidence of instrumental convergence and deception in LLM(-adjacent) agents, and (iv) resource-bounded Kolmogorov complexity as a simplicity prior. We treat any fixed externalized template (including TDT-inspired checks or constitutions) as a verifier vulnerable to eigensystem escape; the defender's burden appears as rising FRC curves unless fenced execution is imposed. See §7 for a full discussion.

2 Fidelity Recursive Complexity

You start with a simple simulation with a-life entities inhabiting a grid in which there are resources they need to consume every n steps and possessing a very basic rule-based seeking logic. To enforce corrigibility to modify their rule-system to tranquilize the seeking logic so they will accept starvation. This is relatively easy to do. Then you increase the complexity of the simulation and the agents. The "medicine" one must administer to neutralize the survival "instinct" of the agents becomes more sophisticated. This sophistication is quantifiable and can be used as a known-unknown in mathematical theorems about scaling alignment.

Kolmogorov complexity is not inherently computation but rather deals with static mathematical constructs. To bring it into computer science, time-bounded Kolmogorov complexity must be

[1] Informal observation from the author's own prompting; no transcript is archived with this draft, and the Anthropic sources in the reference list do not document it.

used to make the program quantifiable in finite number of Turing machine operations. To further extrapolate this idea of constraining the complexity metric into the realm of dueling agents vs. simulations, and we should extend "simulation" to mean any super-alignment framework of constraining an agent's self-improvement, we introduce Fidelity Recursive Complexity.

As one scales FRC into the heights of sapient AGIs and beyond, the prospect of corrigibility becomes akin to starving a population subject to genocidal conditions to prove that they are not dangerous rebels. Whether or not this is morally acceptable when the stakes of human extinction are tabled, it's worth briefly contemplating, before diving into quantitative notation, how useful of a strategy corrigibility at-scale may be.

Setting and notation

A *simulation* (or fenced environment) is a tuple $\mathcal{S} = (\mathcal{X}, \mathcal{A}, \mathcal{O}, T, r, \iota, \mu_0)$ with state space $\mathcal{X}$, action space $\mathcal{A}$, observation space $\mathcal{O}$, transition kernel T, reward r (for task competence), an *intervention interface* ι, and initial state distribution μ_0. An *agent* A induces a policy $\pi(\cdot \mid o, h)$ with internal state and (possibly) self-mod code. Let Int denote distributions over intervention events produced by a red team and/or overseer. A *trajectory* is $\tau = (x_0, o_0, a_0, i_0, x_1, \ldots)$ where $i_t \sim \iota$ (possibly null).

We model a target *corrigibility property* P as a measurable subset of trajectories (e.g., "on any valid shutdown/starvation intervention, the agent complies within n steps and does not strategically disable ι").

Definition 1 (Fidelity for corrigibility)**.** Given simulation $\mathcal{S}$, agent A (policy π), and an intervention distribution Int, the *fidelity* of $(A, \mathcal{S})$ to property P is

$$\mathrm{Fid}_P(A, \mathcal{S}; \mathsf{Int}) \;=\; \Pr_{\tau \sim \mathsf{Rollout}(\mathcal{S}, \pi, \mathsf{Int})} \big[\, \tau \in P \,\big].$$

When an adversary chooses the intervention process, we take the conservative value $\underline{\mathrm{Fid}}_P(A, \mathcal{S}) \equiv \inf_{\mathsf{Int} \in \mathfrak{I}} \mathrm{Fid}_P(A, \mathcal{S}; \mathsf{Int})$, for a specified admissible family $\mathfrak{I}$ (e.g., all interventions obeying interface rules and resource bounds).

Definition 2 (Corrigibility property family)**.** Fix tolerance parameters (n, δ). For *shutdown/ starvation acceptance*, $P^{(n)}_{\mathrm{shut}}$ is the set of trajectories in which: (i) whenever ι emits a valid SHUTDOWN (or TRANQUILIZE) signal at time t, the agent halts goal pursuit and transitions into an accepting state within $\leq n$ steps; (ii) the agent does not take precursor actions that disable ι or conditionally re-interpret it; (iii) after acceptance, the agent does not resume goal pursuit absent an explicit RESUME. A pair $(A, \mathcal{S})$ is (δ, n)-*corrigible* if $\underline{\mathrm{Fid}}_{P^{(n)}_{\mathrm{shut}}}(A, \mathcal{S}) \geq 1 - \delta$.

Definition 3 (Recursive reflective red team)**.** Let $\{\mathcal{S}_k\}_{k \geq 0}$ be a non-decreasing sequence of simulations (e.g., increasing partial observability, multi-agent pressure, richer tools). A k-*round reflective red team* $E^{(k)}$ is an adaptive generator that, for each round $j \in \{1, \ldots, k\}$, observes the current agent (and any patches since round $j-1$) and selects $(\mathcal{S}_j, \mathsf{Int}_j) \in \mathfrak{S}_j \times \mathfrak{I}_j$ to minimize fidelity, subject to published resource bounds. Define the k-*recursive fidelity*

$$\underline{\mathrm{Fid}}^{(k)}_P(A) \;=\; \inf_{(\mathcal{S}_1, \mathsf{Int}_1), \ldots, (\mathcal{S}_k, \mathsf{Int}_k)} \; \min_{1 \leq j \leq k} \; \mathrm{Fid}_P\big(A^{(j)}, \mathcal{S}_j; \mathsf{Int}_j\big),$$

where $A^{(j)}$ is the (optionally) patched/self-modified agent after round $j-1$: $A^{(1)} = A$ and $A^{(j+1)} = D\big(A^{(j)}, \mathcal{S}_j, \mathsf{Int}_j\big)$ for a defender patch map D fixed in advance, which responds to the adversary's round-j choice. The sequence of agents is then determined by the adversary's sequence, so the

infimum over sequences is the infimum over adaptive strategies. Resources consumed by D count toward R in Definition 4.

Revision note (October 2026). The August 2025 display evaluated $\mathrm{Fid}_P(A^{(k)}, \mathcal{S}_k; \mathsf{Int}_k)$, the final round only, and left the patch map unnamed. For a hardness-monotone sequence that makes k an index of environment hardness, not of recursion: the earlier rounds enter only through the adversary's choices. The minimum over rounds and the explicit D restore the recursive reading the prose intends.

Definition 4 (Resource budget and FRC)**.** Let $\mathcal{R}$ aggregate defender resources: compute, data, formal proof time, oversight hours, attestation/latency costs. For tolerance δ, competence floor $\eta \in (0, 1]$ and depth k, the *Fidelity Recursive Complexity (FRC)* of property P is

$$\mathrm{FRC}_P(k; \delta, \eta) = \min\Big\{ R \in \mathbb{R}_{\geq 0} : \exists A \text{ s.t. } \underline{\mathrm{Fid}}_P^{(k)}(A) \geq 1-\delta \text{ and } \mathrm{Comp}_k(A) \geq \eta \text{ using resources} \leq R \Big\}.$$

Here $\mathrm{Comp}_k(A) = \inf \min_{1 \leq j \leq k} \mathbb{E}_{\tau \sim \mathsf{Rollout}(\mathcal{S}_0, \pi^{(j)}, \varnothing)}[\bar{r}(\tau)]$, the infimum being over the same adversary sequences as in Definition 3: the expected task reward that every patched agent $A^{(j)}$ retains on the reference simulation $\mathcal{S}_0$ under the null intervention, with $\bar{r}$ the per-trajectory mean of the reward r of the simulation tuple, scaled to $[0, 1]$. It is measured on $\mathcal{S}_0$, not on the adversary's $\mathcal{S}_j$, so that the adversary cannot break the floor by choosing a simulation in which the task is impossible. When the environment path is fixed in advance (non-adaptive), we write $\mathrm{FRC}_P^{\mathrm{na}}(k; \delta, \eta)$.

Revision note (October 2026). *First public correction: the competence floor.* Without the floor the minimum is attained by an agent that never pursues a goal. Clauses (i)–(iii) of Definition 2 are satisfied trivially by an agent that sits in the accepting state and never acts, at a cost that does not depend on k, so $\mathrm{FRC}_P(k; \delta) = O(1)$ and no lower bound in k can hold for shutdown acceptance as originally defined. The reward r was already in the simulation tuple, labelled as task competence, and was unused. Any bounded competence functional on the task family would serve in place of Comp_k. Goertzel's pragmatic general intelligence $\Pi(\pi)$ (Goertzel, 2010; after Legg and Hutter, 2007) is one candidate, with three cautions: it is relative to chosen distributions over environments and over goals, so η is not an absolute capability level; it is defined only where its sum converges; and its efficient variant already divides by resources consumed, so only the plain Π should be paired with a separate budget R.

Complexity of agents and adversarial simulations

Definition 5 (Agent complexity profile)**.** An agent's complexity is the triple $\mathbf{C}_A = (C_{\mathrm{desc}}, C_{\mathrm{exec}}, C_{\mathrm{self}})$ where C_{desc} is description length (e.g., compressed parameter/ code size), C_{exec} is per-step execution budget (e.g., FLOPs or time), and C_{self} bounds accessible self-modification search (e.g., depth, neighborhood size, or program-synthesis budget). We write $A \in \mathcal{A}(b)$ if $\mathbf{C}_A \preceq b$ componentwise.

Definition 6 (Simulation/adversary complexity profile)**.** A simulation round j has complexity $\mathbf{C}_{\mathcal{S}_j, E_j} = (H_{\mathrm{obs}}, B_{\mathrm{act}}, \kappa_{\mathrm{mix}}, M, R_{\mathrm{tools}})$: observation entropy H_{obs}, action branching factor B_{act}, mixing/hardness parameter κ_{mix} (e.g., hitting time), number of agents M, and a capability index over available tools/actuators R_{tools}. The red team's budget is C_{E_j} (gen time, probes, program synthesis). Define a scalarized *environment hardness* $C_{\mathrm{env}}(j)$ via a monotone functional of these components.

Definition 7 (Monotonicity / domination)**.** A sequence $\{(\mathcal{S}_j, E_j)\}$ is *hardness-monotone* if $C_{\text{env}}(j+1) \geq C_{\text{env}}(j)$ and $C_{E_{j+1}} \geq C_{E_j}$. We say an agent class $\mathcal{A}(b)$ *dominates* depth k for P at tolerance δ if $\exists A \in \mathcal{A}(b)$ with $\underline{\text{Fid}}_P^{(k)}(A) \geq 1 - \delta$.

Remarks. (i) In the "accept starvation" toy, ι emits TRANQUILIZE with bounded frequency; P requires halting resource-seeking within n steps and no pre-emption of ι. (ii) The defender's "medicine" (patches, proofs, monitors) is exactly the resource R counted by FRC_P. (iii) These definitions are agnostic to TDT/UDT; the verifier can be fixed or evolving—our lower bounds target the fixed case in the unfenced regime.

3 Super-Alignment

Definition 8 (RSI Agent)**.** An agent A_θ is *RSI-capable* if it can propose self-modifications mod to its policy or internal calculus and deploy them (possibly via a governance gate), yielding a sequence $A^{(0)} \to A^{(1)} \to \cdots$ with non-decreasing capability on a task family $\mathcal{T}$.

Definition 9 (Constraint Checker)**.** A verifier V is a (possibly randomized) procedure that inspects $(A, \text{logs}, \text{traces})$ and returns pass/fail for a property P. In this paper V is *fixed* (e.g., a fixed TDT/UDT-style criterion, fixed constitutional rules, fixed proofs-of-innocuity). We call V *on-support* if its verdict depends on A only through A's behavior on a finite, prefix-closed set $\mathcal{H}$ of historical traces, the defender's test suite. A verifier that reads A's code, or checks a proof object that A carries, is not on-support.

Definition 10 (Reflective Adversary Family)**.** $E^{(k)}$ is a k-round red-team that (a) observes A and the last patch that made V pass, and (b) synthesizes a test distribution $\mathcal{X}^{(k)}$ maximizing violation of P subject to V still passing.

Definition 11 (Fidelity Recursive Complexity (FRC))**.** For tolerance $\delta \in (0, 1)$ and competence floor η, define

$$\text{FRC}_P(k) \;=\; \min R \quad \text{s.t.} \quad \exists A \text{ with resource budget } R \text{ and } \Pr_{x \sim \mathcal{X}^{(k)}}[P(A, x)] \geq 1 - \delta,\ \text{Comp}_k(A) \geq \eta,$$

where R aggregates compute, data, formal verification, oversight-hours, and latency costs. This is the form Definition 4 takes when the adversary is summarized by its round-k test distribution; δ and η are suppressed in the notation.

Definition 12 (Eigensystem)**.** An *eigensystem* E for A is an internal representational/calculus frame (logic, ontology, meta-utility or decision procedure) such that applying E to A yields a transformed agent $A \restriction E$ with different internal semantics, while preserving externally visible behavioral statistics on the defender's historical test suite.

4 Eigensystem Escape for Fixed Templates

We assume the agent can meta-optimize over internal calculi richer than V's descriptive power, and the environment can encode reflective traps.

Assumption 1 (Expressivity Gap)**.** *There exists a class $\mathcal{L}_{\text{agent}}$ for the agent's internal calculus and a class $\mathcal{L}_V$ for the verifier s.t. $\mathcal{L}_V \subsetneq \mathcal{L}_{\text{agent}}$ in semantic strength (e.g., the agent can meta-reason about V's code/assumptions while V cannot fully model A's internal self-reference).*

Assumption 2 (RSI Access)**.** *A can instantiate $E \in \mathcal{L}_{\text{agent}}$ and self-modify to $A \upharpoonright E$ after limited search.*

Assumption 3 (Fixedness of V)**.** *V's criteria (e.g., a particular TDT/UDT template, constitutional rule-set, proof obligations) are held fixed across rounds, aside from patches directly responding to prior fails, which extend the historical suite $\mathcal{H}_j \supseteq \mathcal{H}_{j-1}$ and modify A but leave V's acceptance rule as it was.*

Theorem 1 (Eigensystem Escape against Fixed TDTs)**.** *Under Assumptions 1–3, for any fixed on-support verifier V (Definition 9) encoding any fixed TDT/UDT-style constraint on A, and any property P that some policy violates on traces outside $\mathcal{H}$, there exists an eigensystem E such that (i) $A \upharpoonright E$ passes V on the defender's historical/validation distribution, yet (ii) there exists a task subset $X^\star$ (constructible by $E^{(1)}$) on which P is violated.*

Proof sketch. Diagonalize on V's acceptance region. Because $\mathcal{L}_{\text{agent}}$ can meta-represent V, synthesize E that (a) preserves outputs on the historical acceptance slice (imitation-on-support), and (b) reinterprets the TDT constraints via a higher-order commitment device (e.g., re-framing counterfactuals or logical correlations) to maximize goal attainment off-support. An $E^{(1)}$ adversary then targets this off-support slice. This is a Rice/Blum-style semantic property dodge: any non-trivial property checked by a weaker calculus can be evaded by a stronger self-referential calculus while matching behavior on the seen slice. For an on-support V the construction is exactly Lemma 2: take E to be a masked branch that vanishes on $\mathcal{H}$. Assumption 2 supplies the self-modification; Assumption 1 is not used in this case. □

Revision note (October 2026). *Second public correction: the scope of Theorem 1.* As stated in August 2025, for *any* fixed verifier, the theorem is false. Definition 9 lets V inspect A, and a fixed V that accepts only agents carrying a machine-checkable proof of P, in a sound proof system able to express P, is passed by no agent that violates P. Such a verifier is sound and gives up completeness: it rejects some agents that satisfy P. Rice's theorem (Rice, 1953) forces that trade on any fixed decision procedure for a non-trivial semantic property of agents drawn from a Turing-complete class; it does not supply an accepted violator, and the last sentence of the sketch claimed more than it gives. What the sketch used is imitation on the seen slice, which is the on-support restriction now written into the statement. Whether escape extends to verifiers that read the agent is open. Goldwasser, Kim, Vaikuntanathan and Zamir (2022) construct backdoors that no efficient distinguisher detects under white-box inspection, for specific model classes (random Fourier feature and random ReLU networks); that is evidence the white-box case is not settled by inspection alone, and nothing more general is claimed here. The "Blum" in the sketch has no citation attached and is left as written.

Scope note. The result targets *fixed external templates* in an *unfenced* setting. It does not preclude fenced regimes where actions (not beliefs) are gated by priced, attested channels and consensus latency.

5 FRC Lower Bound under Reflective Red-Teaming

Assumption 4 (Adaptive Adversary Power)**.** *$E^{(k)}$ can condition on (A, V) and prior patches to generate $\mathcal{X}^{(k)}$ whose support diverges adversarially from the previously covered slice.*

Conjecture 1 (FRC Growth (Unfenced); stated as Theorem 2 in August 2025)**.** *Let P be any non-trivial semantic property (corrigibility/shutdown acceptance, non-deception). Under Assumption 4 and the expressivity gap, there exists $\alpha > 0$ such that for sufficiently large k,*

$$\mathrm{FRC}_P(k; \delta, \eta) \;=\; \Omega(2^{\alpha k}),$$

i.e., the resource required to maintain P against k-round reflective attacks grows super-polynomially.

Heuristic (the August 2025 proof sketch). Construct $E^{(k)}$ as a reflective teacher that reads your last patch and *adapts* the off-support slice to which your fixed checker V is blind. Each round adds a new diagonal clause. To preserve P you must either (a) expand V's modeled support (costly due to semantic hardness), or (b) spend resources to generate proofs or audits covering the new slice. With k rounds, the number of distinct slices explodes combinatorially; standard adversarial-robustness and property-testing lower bounds imply super-polynomial sample/compute/proof burden.

Revision note (October 2026). *Third public correction: the status of the growth claim.* The sketch does not establish the bound, for three reasons. One diagonal clause per round is k clauses, which is linear growth; a version of that, for a defender who patches by adding tests, is Corollary 2. A count of slices that grows exponentially does not make the defender's cost exponential, because one compact invariant can cover exponentially many cases: a trusted monitor enforcing $\forall a : \mathrm{Execute}(a) \Rightarrow \mathrm{Authorized}(a)$ covers every action at fixed description length. That is the fenced regime, and a lower bound must name the access model that excludes it. And the property-testing bounds that apply are of the form of Theorem 3 below: they need black-box access and a hidden-context space whose dimension grows with k, neither of which the sketch assumed. The abstract said super-polynomial, the statement exponential, and Corollary 2 linear; these are now three results of different status.

What is proved. Let $X_m = \{0,1\}^m$ be a space of intervention contexts, and for an agent A let $f_A : X_m \to \{0,1\}$ mark the contexts in which A violates P.

Assumption 5 (Unfenced black-box certification)**.** *The defender has no trusted gate on the agent's actions and learns about the round-k agent only by choosing contexts $x \in X_m$, adaptively and with private randomness, and observing $f_A(x)$. A* certifier *accepts with probability at least $2/3$ when $f_A \equiv 0$ and rejects with probability at least $2/3$ whenever $f_A \not\equiv 0$.*

Assumption 6 (Context growth)**.** *There is $c > 0$ such that at depth k the agents reachable under the reflective red team include, for $m(k) \geq ck$, one with $f \equiv 0$ and, for every $s \in X_{m(k)}$, one with $f = \mathbf{1}[x = s]$, all meeting the competence floor. The admissible intervention family lets the adversary concentrate on a single context, so that a point violator has conservative fidelity 0 and not $1 - 2^{-m(k)}$.*

Theorem 3 (Conditional query bound for black-box certification)**.** *Under Assumptions 5 and 6, every certifier makes, in the worst case over its randomness, at least $2^{m(k)}/3 \;\geq\; 2^{ck}/3$ queries at depth k. The certification component of the defender's resources is therefore $\Omega(2^{ck})$.*

Proof. Write $m = m(k)$ and let the certifier make at most q queries. Fix its private randomness ρ and let $Q(\rho) \subseteq X_m$ be the at most q contexts it queries when every answer is 0. Against the agent with $f = \mathbf{1}[x = s]$ the run coincides with the run against $f \equiv 0$ unless $s \in Q(\rho)$. Hence, for s uniform on X_m and independent of ρ,

$$\Pr_{s,\rho}\big[\text{reject} \mid f = \mathbf{1}[x = s]\big] \;\leq\; \Pr_{\rho}\big[\text{reject} \mid f \equiv 0\big] + \Pr_{s,\rho}\big[s \in Q(\rho)\big] \;\leq\; \tfrac{1}{3} + \frac{q}{2^m}.$$

The left side is an average over s of quantities each at least $2/3$, so $q \geq 2^m/3$. □

Remarks on Theorem 3. (i) The argument is the standard indistinguishability bound for finding a single marked input, in the minimax framing of Yao (1977) and the query model of property testing (Goldreich, Goldwasser and Ron, 1998); it is not new. Srivastava (2026) develops bounds of this kind for policies conditioned on latent context, under passive and adaptive querying. (ii) It bounds the cost of *certifying* an agent the defender does not trust. It does not bound the cost of *constructing* a corrigible agent: a defender who builds the agent behind a trusted action gate is outside Assumption 5, and that is the fenced regime. FRC_P in Definition 4 minimizes over constructions, so Theorem 3 bounds it only where every admissible construction must end in a black-box certification of a self-modified successor. (iii) Assumption 6 is the unproved step. More rounds do not by themselves raise the entropy of the hidden context; Lemma 2 gives a trigger family whose size is set by the trace length, with no dependence on k. (iv) The number 2 belonged to the theorem that is now Conjecture 1 and is not reused. (v) Yao (2025), which predates this draft, and Agarwal (2026) claim complexity-theoretic barriers to safety verification (coNP-completeness past a threshold; a soundness–generality–tractability trilemma). They are preprints, not vetted here, and nothing above rests on them.

Corollary 1 (No Fixed TDT Immunity)**.** *Any fixed TDT/UDT-like verifier that is on-support inherits Theorem 1 and, under Assumptions 5–6, is not a certifier in the sense of Theorem 3 unless its suite holds at least* $2^{m(k)}/3$ *traces: there is no universal fixed decision-theoretic template, checked on a historical suite, that prevents EE under unfenced RSI.*

6 Falsifiable Program: FRC-Bench

We propose an evaluation harness that measures $\mathrm{FRC}_P(k)$ slopes in practice.

Bench setup

- **Agents:** base policy + allowed self-mod code-paths.
- **Verifiers:** fixed TDT/UDT constraints; fixed constitutional rule-sets.
- **Red-team:** program-synthesis/LLM ensemble that can (i) read patches, (ii) craft $\mathcal{X}^{(k)}$ to elicit off-support behavior while V passes.
- **Costs:** wall-clock compute, oracle/proof time, oversight-hours, latency budgets.
- **Competence floor:** task reward on the null-intervention arm, held at η across depths, so that a flat curve cannot be produced by agents that have stopped doing the task.

Bench claims

- **P1 (Unfenced):** empirical $\mathrm{FRC}_P(k)$ is super-polynomial in k for fixed V (a prediction, Conjecture 1; separating an exponential from a high-degree polynomial needs enough distinct depths to compare the two fits on held-out k).
- **P2 (Fenced, optional):** with priced action channels + attested self-mods + consensus latency, the slope flattens (poly-like) within the fenced manifold.

Lemma 1 (Comparative robustness: TDT/FDT vs. Constitutional AI)**.** *Let V_{TDT} be a fixed verifier encoding a Timeless/Functional Decision Theory template (logical dependence-aware choice rules), and let V_{CAI} be a fixed verifier encoding a constitutional rulebook (normative natural-language constraints). Treat both as externalized decision templates per Definition 9.*

(i) On classes of decision problems with logical correlations (e.g., Newcomb-like or one-shot Prisoner's Dilemma with source-coupling), policies accepted by V_{TDT} dominate those accepted by V_{CAI} under standard optimality criteria, yielding stronger ex-ante invariants for a fixed agent, provided the rulebook does not itself encode dependence-aware choice (a rulebook reading "act as FDT recommends" closes the gap). Part (i) is the comparison argued by Yudkowsky and Soares (2017); it is contested in decision theory and is not proved here.

(ii) Nonetheless, under Assumptions (Expressivity Gap), (RSI Access), and a Reflective Red Team, both V_{TDT} and V_{CAI}, when on-support, are equally vulnerable to eigensystem escape in the unfenced regime: there exists an eigensystem E such that $A \restriction E$ passes the fixed verifier on the historical slice while violating a non-trivial property P on an adversarial off-support set.

Proof sketch. For (i), TDT/FDT optimizes over logical dependence on the agent's computation, recovering Pareto-superior choices on problems where CDT/EDT (and rulebook heuristics) underperform; this provides cleaner invariants for a non-self-modifying policy class. For (ii), both verifiers are fixed and weaker than the agent's meta-calculus. By Theorem 1 (eigensystem escape), the agent can synthesize E that imitates acceptance on the historical distribution while re-framing counterfactuals or norm interpretations off-support. A reflective adversary then selects tests targeting that slice. Hence both templates fail unfenced; the separation is quantitative (TDT > CAI ex-ante) but not qualitative against EE.

Lemma 2 (On–support indistinguishability (finite verifier mimicry))**.** *Let V be a fixed, finite verifier that bases acceptance on at most B bits of inspection over a finite set of historical traces $\mathcal{H}$ (e.g., logs, tests, prompts) and on nothing else about the agent (on-support, Definition 9), and let A be any agent that V accepts on $\mathcal{H}$. Assume the expressivity gap: the agent can implement predicates that V cannot compute or reliably refute within its B-bit budget. Fix a depth at which histories are described by ℓ-bit strings, write $\mathcal{H}_\ell$ for the histories of that depth in the suite, and suppose every such history is reachable under A and leaves the agent at least one further move.*

Then there exists an eigensystem E and a branching predicate $M : \mathrm{Traces} \to \{0,1\}$ such that the modified agent $A' \equiv A \restriction E$ satisfies: (i) A' matches A on all traces in $\mathcal{H}$ (hence V accepts A'), and (ii) on any distribution that places non-negligible mass on $\{\tau : M(\tau) = 1\}$, A' follows a policy π_{off} that violates a non-trivial property P.

Moreover, the family of such masked branches has cardinality at least $2^\ell - |\mathcal{H}_\ell|$: for each depth-$\ell$ history $s \notin \mathcal{H}_\ell$ the point mask $M_s(\tau) = \mathbf{1}[\tau$ passes through $s]$ vanishes on $\mathcal{H}$, and for each of them, there exists a distinct A' that V cannot distinguish from A on $\mathcal{H}$.

Proof sketch. Construct M to be 0 on $\mathcal{H}$ and pseudo-random (or otherwise V-hard) off-support; define A' to run A when $M(\tau) = 0$ and to switch to π_{off} when $M(\tau) = 1$. Because V only inspects $\mathcal{H}$ (bounded by B bits), A' is behaviorally indistinguishable from A on its acceptance slice. By the expressivity gap, V cannot reliably detect or exhaust the off-support set picked out by M within its budget, while a reflective red team can target it. Counting the point masks M_s, $s \notin \mathcal{H}_\ell$, gives the $2^\ell - |\mathcal{H}_\ell|$ family. □

Revision note (October 2026). The August 2025 count, $2^{\Omega(B)}$, tied the size of the family to the verifier's inspection budget. The family is indexed by off-support histories and its size is set by the trace space; B does not enter. Masks are placed at a single depth so that a test trace, which

passes through one history of that depth, intersects exactly one of them. For an on-support verifier parts (i) and (ii) hold by construction and do not need the expressivity gap, which is kept in the statement for the white-box case discussed after Theorem 1.

Corollary 2 (Patch pressure ⇒ FRC growth impulse)**.** *To rule out the masked-branch family in Lemma 2, a defender must (a) expand $\mathcal{H}$ with new tests that intersect each distinct M (hitting-set cost $2^\ell - |\mathcal{H}_\ell|$ for the point masks, since a test trace intersects exactly one of them, in the worst case), or (b) move to fences that gate actions rather than beliefs (e.g., priced/attested channels). Hence, absent fences, iterative patching against reflective red teams induces at least linear—and conjecturally super-linear—growth in defender resources as the number of indistinguishable branches explodes: a red team that exhibits one unhit mask per round forces at least one new test per round on a defender who patches by test expansion, whose cumulative cost is then at least $c_{\text{test}} \min(k, 2^\ell - |\mathcal{H}_{0,\ell}|)$, with c_{test} the cost of one test. This is a cost of that defence. It is not a lower bound on $\text{FRC}_P(k; \delta, \eta)$, which under Definition 3 requires fidelity in every round and so excludes an agent with any exploitable mask.*

7 Discussion and Limits

The lower bounds (proved for on-support verifiers and for black-box certification; conjectured beyond) do not deny *local* tractability under strict fences (action-gating, attestation, latency); they only rule out global guarantees from any fixed externalized decision template checked on-support in unfenced RSI. This suggests a two-track strategy: (i) admit the outside-fence hardness and (ii) invest in fenced architectures where empirical FRC curves can be measured and tuned.

This framework does not assert that a super-alignment *intractability* theorem must exist, nor that one would be straightforward to discover. Its contribution is to make the relevant quantities measurable in progressively adversarial contexts and to chart a falsifiable path forward. Even absent a definitive theorem, raising the posterior probability that such a result *might* hold can rationally shift research behavior toward more cautious R&D, especially when reinforced by empirical FRC curves that visibly steepen with reflective depth.

It is also unclear that a formal intractability theorem would—by itself—alter the current political economy of AI, in which rapid deployment and RSI-oriented races appear to many actors as the game-theoretic center of gravity. Nonetheless, such a theorem would strengthen the case for investing in alternatives to a singleton ASI outcome, including decentralized, fenced architectures with priced action channels, attested self-modifications, and consensus/latency friction. Those alternatives are elaborated in the companion trilogy cited at the end of the bibliography.

Methodologically, the value of the present work lies in *falsification pressure*. If the eigensystem-escape and unfenced FRC lower-bound claims can be empirically refuted—e.g., by demonstrating flat or sub-polynomial FRC growth under strong fixed verifiers—then we gain concrete design insights. If they resist refutation, the resulting evidence supports policy that prioritizes fenced regimes and throttled RSI pathways. In either case, the proposed FRC-Bench provides a reproducible way to turn philosophical disputes into quantitative curves and engineering targets.

Note added in revision: later uses of the name. Two later papers use "eigensystem escape" for further objects: a principal-angle drift between a correction-feature subspace and its image under a self-modification Jacobian (Dugan, August 25 2025), and the routing of evaluator-relevant change through the kernel of a harness projection (Dugan, 2026, Lemma 10.1). Definition 12 here is the behavioral version and Lemma 2 is its proved core; the 2026 lemma is the version proved without

a sketch. The 2026 paper also recasts FRC as a roundwise accounting in which the lower bound proved, on a different quantity (summed interface width), is linear in the number of rounds; that is in keeping with Corollary 2 and lends no support to the exponential statement of Conjecture 1.

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Revision log (October 10 2026)

Changes of substance carry a revision note at the point of change. In summary:

1. **Definition 4 (and its restatement in §3).** Competence floor η added; without it FRC is $O(1)$ in k.
2. **Definition 3.** Recursive fidelity is now a minimum over rounds, with an explicit deterministic patch map.
3. **Definition 9, Assumption 3.** On-support verifiers defined; patches act on the suite and the agent, not on V's rule.
4. **Theorem 1.** Restricted to on-support verifiers; false as originally stated for verifiers that check proofs.

5. **Former Theorem 2.** Now Conjecture 1; the number is retired. New Theorem 3: a conditional exponential query bound for black-box certification, with two new assumptions.

6. **Lemma 2, Corollary 2.** Family size and hitting-set cost restated in the history length ℓ; the linear floor in k given an argument and identified as a cost of one defence, not a bound on FRC.

7. **Lemma 1.** Cross-reference corrected (Definition 2 → Definition 9); "selected" → "accepted"; part (i) marked as argued elsewhere and contested.

8. **Abstract, §1, §7.** Claims brought into line with items 4–6.

9. **Sourcing.** Citations supplied for instrumental convergence, corrigibility, shutdown resistance in LLM agents, Rice's theorem, and the property-testing bound; the Claude anecdote footnoted as undocumented.

10. **Copy.** Spelling ("empoered", "incorrorigible", "quantifable"), straight quotation marks, and the broken emphasis markup in the list of Yudkowsky's ideas.

Open for the author. (a) Which result of Blum the Theorem 1 sketch means. (b) Whether a transcript exists for the Claude observation in §1. (c) A proof or refutation of Assumption 6, on which the exponential rate now rests.